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Data Article

A Congressional Twitter network dataset quantifying pairwise probability of influence

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ABSTRACT

We present a social network dataset based on interactions between members of the 117th United States Congress between Feb. 9, 2022, and June 9, 2022. The dataset takes the form of a directed, weighted network in which the edge weights are empirically obtained "probabilities of influence" between all pairs of Congresspeople. Twitter's application programming interface (API) V2 was used to determine the number of times each member of Congress retweeted, quote tweeted, replied to, or mentioned other Congressional members, and the probability of influence was found by normalizing the summed influence by the number of tweets issued by each Congressperson. This network may be of particular interest to the study of information diffusion within social networks.

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Specifications Table

Subject	Data Science: Data Mining and Statistical Analysis
Specific subject area	Information diffusion on complex networks
Data format	Raw, Analyzed
Type of data	Social network connectivity
Data collection	We used the Tweepy Python package to interact with Twitter's API, obtaining the most recent 3,200 tweets of all members of the 117th United States Congress. From this initial tweet dataset, we excluded tweets that were created outside of a common time window. We also excluded any Congressional member who tweeted less than 100 times during this window of time. We then used the remaining tweets to quantify the probability of one user influencing another (through retweets, quote tweets, replies, or mentions).
Data source location	All tweets were obtained through the Twitter API on June 9, 2022.
Data accessibility	Repository name: Zenodo DOI: https://doi.org/10.5281/zenodo.8253486 Direct URL: https://zenodo.org/record/8253486 Original network connectivity data also shared on the Stanford Network Analysis Project (SNAP) repository: https://snap.stanford.edu/data/congress-twitter.html Data Use Agreement: https://developer.twitter.com/en/developer-terms/agreement-and-policy
Related research article	C.G. Fink, K. Fullin, G. Gutierrez, N. Omodt, S. Zinnecker, G. Sprint, S. McCulloch, A centrality measure for quantifying spread on weighted, directed networks, *Physica A*, Vol. 626 (2023). [1]

1. Value of the Data

- This dataset is useful because it is, to our knowledge, the only publicly available Twitter dataset that empirically quantifies pairwise influence between users. The closest we are aware of is the Higgs Twitter dataset [2], which tracked Twitter posts about the Higgs boson in the week of its discovery. This dataset includes information about re-tweets, replies, and mentions, but it does not synthesize these data into influence probabilities. Furthermore, most users in the dataset only tweeted a few times, making the inference of empirical probabilities problematic.
- Anyone interested in analyzing complex networks, especially social networks, will benefit from this dataset.
- This dataset can be used to further investigate information diffusion in social networks and could also be useful in the study of epidemiological models. There is a vast literature on identifying 'influential spreaders' within social or epidemiological networks [3], with much work focused on developing centrality measures to more accurately and efficiently perform such identification [4]. These studies typically propose a novel centrality measure and evaluate its performance on various real-world networks. The centrality measure's performance is often compared with "ground truth" results from Monte Carlo simulations of a simple epidemiological model (such as the Susceptible-Infected-Recovered, or SIR, model). Since (to our knowledge) no previous network datasets have quantified connection-by-connection transmission probabilities, these Monte Carlo simulations typically just *assume* a homogeneous transmission probability, set to an arbitrarily chosen value. Our dataset's quantification of transmission probabilities may therefore be valuable to researchers who are interested in developing centrality measures specifically for networks in which transmission probabilities are not homogenous.
- This dataset may also be useful to political scientists interested in social network interactions between members of the 117th United States Congress.

2. Data Description

All data files and code are hosted on the Zenodo repository linked above.

Data files corresponding to data used in the research paper (adhering to Twitter's Developer Agreement and Policy):

- `congress_ids.txt`: contains 530 Twitter user IDs for each member of the 117th United States Congress with active Twitter accounts on June 9th, 2022
- `congress_tweet_ids.txt`: contains 117,974 Twitter tweet IDs for all tweets authored by the users with IDs in congress_ids.txt from February 9, 2022 to June 9, 2022
- `congress_network_data.json`: a JSON file containing the following Twitter Influence Network data:
  - `inList`: list of lists such that `inList[i]` is a list of all nodes sending connections TO node i
  - `inWeight`: list of lists containing the connection weights (transmission probabilities) corresponding to the connections in `inList`
  - `outList`: list of lists such that `outList[i]` is a list of all nodes receiving connections FROM node i
  - `outWeight`: list of lists containing the connection weights (transmission probabilities) corresponding to the connections in `outList`
  - `usernameList[i]` gives the Twitter username corresponding to node i
- `congress.edgelist` contains the weighted, directed edgelist for the Twitter Influence Network, in [NetworkX edgelist] (https://networkx.org/documentation/stable/reference/readwrite/edgelist.html) format

Code files for using the Twitter API to fetch the data, making a "Twitter Influence Network" graph, and running the Viral Centrality measure algorithm over the graph:

- `tweet_rehydrater.py`: code for rehydrating tweet IDs
  - Makes use of utility variables and functions in `utils.py`
  - See Twitter's More About Restricted Uses of the Twitter APIs webpage for more information about rehydration
- `tweet_fetcher.py`: code for fetching tweets from a user ID
  - Makes use of utility variables and functions in `utils.py`
- `graph_maker.py`: code for making a Twitter social network graph from rehydrated or fetched tweets
- `test_graph_maker.py`: unit test for graph_maker.py. This file helps explain how influence is modeled in a Twitter social network graph
- `compute_vc.py`: code to implement the Viral Centrality portion of Fig. 2A from "A centrality measure for quantifying spread on weighted, directed networks" (uses the function in `viral_centrality.py`)
- `histogram_weights.py`: code to replicate Fig. 2B from "A centrality measure for quantifying spread on weighted, directed networks"

3. Experimental Design, Materials and Methods

On June 9, 2022, we used Twitter's API to obtain the most recent 3,200 tweets (the maximum number allowed by Twitter using their `/2/users/:id/tweets` API endpoint) of all members of the 117th United States Congress with active Twitter accounts (530 Congressional members out of 535). To do this, we first obtained the relevant Congressional Twitter handles from this URL: https://pressgallery.house.gov/member-data/members-official-twitter-handles. Using the `/2/users/by` API endpoint, we were able to lookup the unique Twitter user ID for each handle. These user IDs are provided in the `congress_ids.txt` file described previously in the Data Description section. Next, we used the `/2/users/:id/tweets` endpoint to fetch

recent tweets for each user ID. We parameterized our tweet requests to ask for each tweet's referenced tweets via the `referenced_tweets` and `referenced_tweets.id.author_id parameters`, its replied to user ID via the `in_reply_to_user_id parameter`, and its mentions via the entities parameter. We then parsed the API responses, extracting each tweet's ID and, if available, information about this tweet's interaction with other users. This interaction information included the user ID this tweet was replying to, the user IDs mentioned in the tweet text, and the user IDs who authored tweets this tweet retweeted or quoted. The tweet IDs are provided in the `congress_tweet_ids.txt` file described previously in the Data Description section. Finally, it was important that we consider only a time interval for which we had data for all Congressional members, so we went as far back in time as possible before first exhausting the 3,200-tweet limit for any one Congressional member. This resulted in our considering the tweets by all Congressional members from February 9, 2022 to June 9, 2022 (179,974 tweets from 525 Twitter accounts). Furthermore, in order for probabilities of influence to be meaningful, we required that a given Congressional member issue at least 100 tweets over this timeframe in order to be included in this dataset. This resulted in 475 Congressional members with tweets included in the social network.

Adhering to the Twitter Developer Agreement and Policy, we have made the Twitter user IDs and tweets from these users from February 9, 2022 to June 9, 2022 available, along with the code necessary to use the Twitter API to "rehydrate" these tweets (i.e. request the full Tweet's content, more information about tweet rehydration is available on Twitter's More About Restricted Uses of the Twitter APIs webpage). Our Twitter API rehydration code is available in a Zenodo repository at this URL: https://zenodo.org/record/8253486. To rehydrate tweets, our code requests, parses, and stores each tweet's retweeted author IDs, quoted author IDs, replied to author IDs, and mention user IDs. With these IDs for each tweet, influence between two members of Congress can be computed. To quantify influence of member i on member j, we considered the number of times member j retweeted, quoted tweeted, replied to, or mentioned a tweet by member i. We then summed all these instances of individual influence and divided by the total number of tweets issued by member i, yielding the empirically estimated probability that any one of member i's tweets would measurably influence member j. Code that calculates influence using this technique and produces a directed, weighted network given tweets IDs is available at the repository URL provided above.

We have also included code for applying our novel Viral Centrality algorithm [1] to this dataset. Viral Centrality efficiently approximates Monte Carlo simulations of the Independent Cascade Model (ICM), in which one "seed node" in a network starts out in an "activated" state and the rest begin in an "inactive" state. On the following time step, each of the neighbors of the seed node is randomly activated (or not) according to the probabilities of influence described above. After a node has been activated, it enters a permanent "recovered" state on the following time step. Each subsequent time step follows the same procedure, with all activated nodes activating inactivated neighbors according to the prescribed probabilities of influence. Simulation of the ICM proceeds for either a fixed number of time steps, or until no more activated nodes remain. A given seed node's influence is quantified according to the average number (across many Monte Carlo simulations) of other nodes it ultimately activates. Our Viral Centrality algorithm provides very good approximation to ICM results without the need for extensive Monte Carlo simulation, drastically cutting down on computation time.

4. Limitations

It is worth noting that tweets returned by the Twitter API may be different over time for several reasons, such as a Twitter user's ability to edit/delete a tweet or change the visibility of their account (e.g., change to private or delete). Consequently, a limitation of this dataset is that the network represents influence on Twitter for 475 members of Congress from Feb. 9, 2022 to June 9, 2022 at the specific moment in time the data were collected (June 9, 2022). The network does not represent Twitter influence during this particular four-month window in its present

state on Twitter, although running the `tweet_rehydrater.py` code file presented in the Data Description section above can produce an influence network for the present state of Tweets from these Congress members during the four-month window. Furthermore, it should be understood that the probabilities of influence which constitute edge weights are empirical estimates taken from a particular window of time, and may not represent their statistically stationary values.

Ethics Statement

All of Twitter's redistribution policies were complied with in developing this dataset. Regarding matching information with specific users, it reads:

...absent a person's express opt-in consent you may only attempt to match your records about someone to a Twitter identity based on:

- **Information provided directly to you by the person.** Note that records about individuals with whom you have no prior relationship, including data about individuals obtained from third parties, do not meet this standard; and/or
- **Public data.** "Public data" in this context refers to:
  - Information about a person that you obtained from a public, generally-available resource (such as a directory of members of a professional association)
  - Information on Twitter about a person that is publicly available, including:
    - Tweets
    - Profile information, including an account bio and publicly-stated location
    - Display name and @handle

Given that the Twitter handles of U.S. Congresspeople were obtained from a public, generally-available source (https://pressgallery.house.gov/member-data/members-official-twitter-handles), Twitter's policy allows for the disclosure of their Twitter handles in our dataset.

Regarding content redistribution, it reads:

The best place to get Twitter Content is directly from Twitter. Consequently, we restrict the redistribution of Twitter Content to third parties. If you provide Twitter Content to third parties, including downloadable datasets or via an API, you may only distribute Tweet IDs, Direct Message IDs, and/or User IDs (except as described below). We also grant special permissions to academic researchers sharing Tweet IDs and User IDs for non-commercial research purposes.

In total, you may not distribute more than 1,500,000 Tweet IDs to any entity (inclusive of multiple individuals associated with a single entity) within any 30 day period unless you have received written permission from Twitter.

Following Twitter's policy, our published Twitter data only includes user IDs and tweet IDs (179,974). For reproducibility, we provide code to rehydrate these tweet IDs, using the Twitter API to request the additional tweet information that is needed to produce a social influence network.

Data Availability

Congressional Twitter Network (Original data) (Zenodo).

CRediT Author Statement

Christian G. Fink: Conceptualization, Methodology, Software, Validation, Writing – original draft, Supervision, Funding acquisition; **Nathan Omodt:** Software, Validation; **Sydney Zinnecker:** Software, Validation; **Gina Sprint:** Conceptualization, Methodology, Software, Validation, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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